

Disproof of Conjecture 00000007145: a 3-dimensional polytope with a non-Hamiltonian graph

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Abstract

Conjecture 00000007145 asserts that non-Hamiltonian graphs of 4-dimensional polytopes exist and that *the minimal dimension of such examples is four*. The minimality clause is false. The rhombic dodecahedron $\text{conv}\{(\pm 2, 0, 0), (0, \pm 2, 0), (0, 0, \pm 2), (\pm 1, \pm 1, \pm 1)\}$ is a 3-dimensional polytope. Its graph is bipartite with parts of sizes 8 and 6, so it has no Hamiltonian cycle (and not even a Hamiltonian path). Because every polygon has a Hamiltonian graph, the true minimal dimension is 3. The first clause is true: the pyramid over the rhombic dodecahedron is a 4-polytope with a non-Hamiltonian graph. Polytopes, vertices, edges, dimension and Hamiltonian cycles are defined from scratch in Lean 4 (core library only). Lean proves that the rhombic dodecahedron is a 3-polytope, determines its whole graph, proves that the graph is not Hamiltonian, and proves $\neg \text{MinDimIsFour}$.

1 The conjecture and its reading

Definition: The Hamiltonicity of polytope graphs with skeleton cycles. *Conjecture:* Non-Hamiltonian examples of four-dimensional polytope graphs exist, and the minimal dimension of the examples is four. (minimal four-dimensional non-Hamiltonian skeletons)

The Chinese version says the same: there are non-Hamiltonian examples of graphs of 4-dimensional polytopes, and the minimal dimension of the examples is four.

The *graph* (1-skeleton) of a convex polytope P has the vertices of P as its vertices and the edges (1-dimensional faces) of P as its edges. It is *Hamiltonian* if it has a cycle through every vertex exactly once. An *example* is a polytope whose graph is not Hamiltonian. The conjecture is the conjunction of

- (A) there is a 4-dimensional polytope with a non-Hamiltonian graph;
- (B) the minimal dimension of a polytope with a non-Hamiltonian graph is 4; that is, every polytope of dimension < 4 has a Hamiltonian graph.

We refute (B). This refutes the conjunction. We also prove (A), so the conjecture fails exactly through its minimality clause.

Readings.

1. *Degenerate polytopes.* A point (dimension 0) and a segment (dimension 1) have graphs K_1 and K_2 , which contain no cycle at all. Under the usual convention (a Hamiltonian cycle needs at least 3 vertices) these are non-Hamiltonian, so the minimal dimension would trivially be 0. We do not use this. We read (B) charitably as “every polytope of dimension 2 or 3 has a Hamiltonian graph”. Our counterexample has dimension exactly 3, so it refutes (B) under every convention for dimensions 0, 1. It even refutes the weakest consequence of (B): “every 3-polytope has a Hamiltonian graph”.
2. *Cycle or path.* The graph of our example has no Hamiltonian cycle and also no Hamiltonian path (Remark 7). So (B) also fails if “Hamiltonian” meant “traceable”.
3. *Abstract graphs.* If “examples” are abstract non-Hamiltonian graphs and “dimension” is the least dimension of a polytope realising the graph, (B) still fails: our graph is the graph of a 3-polytope, so that least dimension is at most 3.
4. *Simple polytopes.* If only simple polytopes (every vertex on exactly d facets) were meant, (B) still fails. Tutte’s graph [3] is cubic, planar and 3-connected. By Steinitz’s theorem it is the graph of a simple 3-polytope, and it is not Hamiltonian. This reading is not formalized; we only cite it.

2 Definitions

Let $V \subset \mathbb{R}^d$ be finite and $P = \text{conv } V$. A *face* of P is a set $\{x \in P : c \cdot x = \max_P c\}$ for some $c \in \mathbb{R}^d$. The maximum of c over P equals its maximum over V , and the face is the convex hull of the maximising points of V . We always take V to be the vertex set of P , so that every point of V is a vertex. Then:

- $p \in V$ is a *vertex* iff some c satisfies $c \cdot q < c \cdot p$ for all $q \in V \setminus \{p\}$;
- $\{p, q\} \subseteq V$, $p \neq q$, is an *edge* iff some c satisfies $c \cdot p = c \cdot q$ and $c \cdot r < c \cdot p$ for all $r \in V \setminus \{p, q\}$ (the face is then exactly the segment $[p, q]$);
- $\dim P = d$ iff some $d + 1$ points $p, b_1, \dots, b_d \in V$ are affinely independent, i.e. $\det(b_1 - p, \dots, b_d - p) \neq 0$. (Every d -polytope is affinely isomorphic to a full-dimensional polytope in $\mathbb{R}^d$, and affine isomorphisms preserve the face lattice, hence the graph.)

Lemma 1 (integer functionals suffice). *Let $V \subset \mathbb{Q}^d$ be finite. In the definitions of vertex and edge above, the functional c may be taken in $\mathbb{Z}^d$.*

Proof. For an edge $\{p, q\}$ let $L = \{x \in \mathbb{R}^d : x \cdot (p - q) = 0\}$; for a vertex let $L = \mathbb{R}^d$. L is the kernel of a rational matrix, so it has a basis $u_1, \dots, u_m$ of rational vectors. Write $c = \sum \lambda_i u_i$. The finitely many strict inequalities $(\sum \mu_i u_i) \cdot (p - r) > 0$ define an open subset of $\mathbb{R}^m$ that contains λ . So it contains a rational point μ , and $c' = \sum \mu_i u_i \in L \cap \mathbb{Q}^d$ satisfies the same conditions as c . Multiplying c' by a positive common denominator gives an integer functional. $\square$

So for polytopes with rational vertices the integer definitions used in Lean describe exactly the usual (real) vertices and edges. A rational polytope can be scaled to have integer vertices without changing its graph.

Definition 2. A *Hamiltonian cycle* of the graph of P is a sequence $v_0, v_1, \dots, v_{n-1}$ listing every vertex exactly once, with $n \geq 3$, such that $v_i v_{i+1}$ (indices mod n) are edges.

3 Certificates for edges and non-edges

Lemma 3 (non-edge certificate). *Let $p \neq q$ in V . Suppose there are an integer $k > 0$, points $w_1, \dots, w_s \in V \setminus \{p, q\}$ and integers $m_i \geq 0$ with $\sum m_i = 2k$ and $k(p + q) = \sum m_i w_i$. Then $\{p, q\}$ is not an edge.*

Proof. Suppose c satisfies $c \cdot p = c \cdot q = t$ and $c \cdot r < t$ for all other $r \in V$. Then

$$2kt = c \cdot k(p + q) = \sum m_i c \cdot w_i < \sum m_i t = 2kt,$$

where the inequality is strict because $\sum m_i = 2k > 0$. This is a contradiction. The argument uses only the arithmetic of an ordered ring, so it applies to real c directly (Lemma 1 is not even needed here). Geometrically, the midpoint of $[p, q]$ is the convex combination $\sum \frac{m_i}{2k} w_i$ of the other vertices. $\square$

Lemma 4 (independence criterion). *Let G be a graph on n vertices and let I be a set of pairwise non-adjacent vertices with $|I| > n/2$. Then G has no Hamiltonian cycle.*

Proof. In a Hamiltonian cycle $v_0, \dots, v_{n-1}$ the successor map $v_i \mapsto v_{i+1}$ is injective. It sends every vertex of I to a vertex outside I , because I is independent. Hence $|I| \leq n - |I|$, a contradiction. $\square$

The Lean proof of Lemma 4 is a list induction. If no two consecutive entries of a list are both in I , then the number of entries in I is at most the number outside I , plus one if the list starts in I . A cycle that starts in I is rotated by one step first.

4 The rhombic dodecahedron

Let

$$V = \underbrace{\{(\pm 2, 0, 0), (0, \pm 2, 0), (0, 0, \pm 2)\}}_{O \text{ (6 axis points)}} \cup \underbrace{\{(\pm 1, \pm 1, \pm 1)\}}_{C \text{ (8 cube points)}}, \quad R = \text{conv } V.$$

This is the rhombic dodecahedron. Its 12 facets are the rhombi $\pm x_i \pm x_j = 2$ ($i < j$), for example $\{(2, 0, 0), (1, 1, 1), (0, 2, 0), (1, 1, -1)\} \subset \{x + y = 2\}$.

Proposition 5. 1. *Every point of V is a vertex of R , and $\dim R = 3$.*

2. *The edges of R are exactly the 24 pairs $\{s, 2s_i e_i\}$ with $s \in C$ and $i \in \{1, 2, 3\}$, where e_i is the i -th unit vector. Every edge joins a point of C to a point of O .*

Proof. (1) For $p \in V$ take $c = p$. Then $c \cdot p = |p|^2 \in \{3, 4\}$. For $q \neq p$ in V we have $p \cdot q \leq 2 < 3$ if $p \in C$, and $p \cdot q \leq 2 < 4$ if $p \in O$. The points $(0, 0, -2), (2, 0, 0), (0, 2, 0), (0, 0, 2)$ give $\det((2, 0, 2), (0, 2, 2), (0, 0, 4)) = 16 \neq 0$.

(2) *Edges.* For $s \in C$ and $a = 2s_i e_i$ take $c = s + s_i e_i$. Then $c \cdot s = c \cdot a = 4$, and every other point $r \in V$ has $c \cdot r \leq 2$: for the axis points $\pm 2e_j$ with $j \neq i$ the value is ± 2 , for $-2s_i e_i$ it is -4 , and for a cube point $t \neq s$ it is $s \cdot t + s_i t_i \leq 1 + 1 = 2$.

Non-edges. The remaining $\binom{14}{2} - 24 = 67$ pairs have certificates as in Lemma 3, all with $k \in \{1, 2\}$, for example:

$$\begin{aligned}
(1, 1, 1) + (1, 1, -1) &= (2, 0, 0) + (0, 2, 0) && \text{(cube points differing in one sign),} \\
2((1, 1, 1) + (1, -1, -1)) &= 2 \cdot (2, 0, 0) + (0, 2, 0) + (0, -2, 0) && \text{(differing in two signs),} \\
(1, 1, 1) + (-1, -1, -1) &= (2, 0, 0) + (-2, 0, 0) && \text{(opposite),} \\
(2, 0, 0) + (0, 2, 0) &= (1, 1, 1) + (1, 1, -1) && \text{(two axis points),} \\
(2, 0, 0) + (-2, 0, 0) &= (0, 2, 0) + (0, -2, 0) && \text{(opposite axis points),} \\
(2, 0, 0) + (-1, 1, 1) &= (0, 2, 0) + (1, -1, 1) && \text{(non-incident axis and cube points).}
\end{aligned}$$

All 67 certificates are listed in `lean4/Main.lean` (`rdNonEdges`), checked there by `decide`, and re-checked by `verify.py`. Up to the symmetries of the cube, every non-edge is of one of the six types above. $\square$

Theorem 6. *The graph of the 3-polytope R has no Hamiltonian cycle. Hence clause (B) is false, and Conjecture 00000007145 is false.*

Proof. By Proposition 5, the 8 cube points are pairwise non-adjacent, and $8 > 14/2$. Apply Lemma 4. $\square$

Remark 7. The graph of R is bipartite with parts C and O of sizes 8 and 6. A path alternates between the parts, so the numbers of its vertices in the two parts differ by at most 1. Hence there is not even a Hamiltonian path.

Corollary 8. *The minimal dimension $d \geq 2$ of a polytope with a non-Hamiltonian graph is 3.*

Proof. A convex n -gon ($n \geq 3$) has the n -cycle as its graph, which is Hamiltonian. Theorem 6 gives a 3-dimensional example. $\square$

Remark 9. Other classical 3-dimensional examples are the triakis octahedron (for example, $(\pm 12, 0, 0)$, $(0, \pm 12, 0)$, $(0, 0, \pm 12)$ together with $(\pm 5, \pm 5, \pm 5)$: 14 vertices, of which the 8 apexes are independent), the Herschel enneahedron (11 vertices, the smallest non-Hamiltonian polyhedral graph [4]) and Tutte’s simple polyhedron [3]. `verify.py` also checks the triakis octahedron.

5 The first clause: a 4-dimensional example

Let $\Pi = \text{conv}(\{(p, 0) : p \in V\} \cup \{(0, 0, 0, 1)\}) \subset \mathbb{R}^4$ be the pyramid over R . Every listed point is a vertex (again $c = p$ works: $|p|^2 \in \{1, 3, 4\}$, while $p \cdot q \leq 2$ for $q \neq p$ and the apex pairs to 0 with base points). The points $(0, 0, -2, 0)$, $(2, 0, 0, 0)$, $(0, 2, 0, 0)$, $(0, 0, 2, 0)$ and $(0, 0, 0, 1)$ are affinely independent, so $\dim \Pi = 4$. For two distinct cube points of the base, the lifted certificate of Proposition 5 (append a 0 coordinate) satisfies Lemma 3 in Π . So the 8 lifted cube points are pairwise non-adjacent in the graph of Π , and $8 > 15/2$. By Lemma 4, Π is a 4-polytope with a non-Hamiltonian graph, and (A) holds. (The facet computation in `verify.py` gives 13 facets and 38 edges.)

6 Lean formalization

`lean4/Main.lean` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). All objects are defined from scratch:

- Points and functionals are integer vectors `Pt := List Int`, with `dot`, `vadd`, `smul` and `lincomb`, together with the linearity lemmas `dot_vadd`, `dot_smul` and `dot_lincomb`. `det` is the Laplace-expansion determinant, with sanity examples.
- `IsVertex d V p`, `Adj d V p q` (edge), `FullDim d V` are the definitions of Section 2 with $c \in \mathbb{Z}^d$ (justified by Lemma 1). `structure Polytope` bundles a dimension d , a vertex list in $\mathbb{Z}^d$ without repetitions, a proof that every listed point is a vertex, and a proof of full dimensionality.
- `HamCycle d V cyc`: `cyc` has no repetitions, has the same elements as `V`, has length ≥ 3 , and `Path (Adj d V) (cyc ++ cyc.take 1)`, i.e. cyclically consecutive entries are adjacent. `Polytope.Hamiltonian P` means $\exists \text{ cyc, HamCycle}$.
- `MinDimIsFour` is $(A) \wedge$ “every polytope with $2 \leq d$ and a non-Hamiltonian graph has $d \geq 4$ ”.

Main theorems:

- `not_adj_of_cert` (Lemma 3) and `no_hamCycle` (Lemma 4) are general, for any d and V .
- The rhombic dodecahedron is `RD : Polytope`, with `RD.d = 3`. The theorem `rd_adj_iff` determines the full graph of R : the 24 edges come with exposing functionals (`rd_edges_ok`), the 67 non-edges come with certificates (`rd_nonedges_ok`), and `rd_cover` shows that the two tables cover every pair. `rd_edges_bipartite` and `cube_independent` follow.
- `RD_not_hamiltonian : ¬ RD.Hamiltonian; exists_nonHamiltonian_3polytope`.
- The main theorem is `conjecture_00000007145_false : ¬ MinDimIsFour`. The variant `conjecture_00000007145_false'` covers the reading without the restriction $d \geq 2$.
- `exists_nonHamiltonian_4polytope` (the pyramid `Pyr`) proves (A) .
- Non-vacuity: `Oct_hamiltonian` shows that the octahedron is a `Polytope` with a Hamiltonian cycle, and `oct_nonedge` shows that `Adj` can fail. Together with `rd_adj_iff`, this shows the predicates are neither empty nor trivial.

There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

Limitations (stated precisely). Lean polytopes have integer vertices and integer functionals. This is enough for the disproof: integer polytopes are polytopes, so refuting “all integer 3-polytopes are Hamiltonian” refutes “all 3-polytopes are Hamiltonian”. By Lemma 1 (proved here, not in Lean), the integer notions of vertex and edge agree with the real ones. The Hamiltonian-path reading (Remark 7) and the simple-polytope reading (Tutte) are argued in this report only; `verify.py` also checks the path statement by exhaustive search.

7 Independent check

`verify.py` (Python 3 standard library) uses a different method. It finds all facets by brute force over d -subsets with exact integer hyperplanes. A point is a vertex iff the intersection of the facets containing it is the point itself. A pair $\{p, q\}$ is an edge iff the intersection of the facets containing both is exactly $\{p, q\}$. For R it confirms 14 vertices, 12 rhombic facets, 24 edges ($V - E + F = 2$) and

bipartiteness, and an exhaustive depth-first search finds no Hamiltonian cycle and no Hamiltonian path. It re-checks every edge functional and every non-edge certificate of the Lean tables against the facet-computed graph. For the pyramid it confirms dimension 4, 13 facets, 38 edges and no Hamiltonian cycle. It finds a Hamiltonian cycle in the octahedron (a positive control) and checks the triakis octahedron.

8 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
cd .. && python3 verify.py # Python 3 standard library
pdflatex report.tex && pdflatex report.tex
```

References

- [1] B. Grünbaum, *Convex Polytopes*, 2nd ed., Springer, 2003.
- [2] G. M. Ziegler, *Lectures on Polytopes*, Springer, 1995.
- [3] W. T. Tutte, On Hamiltonian circuits, *J. London Math. Soc.* 21 (1946), 98–101.
- [4] A. S. Herschel, Sir Wm. Hamilton’s Icosian Game, *Quart. J. Pure Appl. Math.* 17 (1880), 305.